

AP[®] Physics 1: Algebra-Based Practice Exam

From the 2016 Administration

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AP[®] PHYSICS 1 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N·m ² /C ²
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Universal gravitational constant, $G = 6.67 \times 10^{-11}$ m ³ /kg·s ²
Speed of light, $c = 3.00 \times 10^8$ m/s	Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²

UNIT SYMBOLS	meter, m	kelvin, K	watt, W	degree Celsius, °C
	kilogram, kg	hertz, Hz	coulomb, C	
	second, s	newton, N	volt, V	
	ampere, A	joule, J	ohm, Ω	

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam.

- I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- II. Assume air resistance is negligible unless otherwise stated.
- III. In all situations, positive work is defined as work done on a system.
- IV. The direction of current is conventional current: the direction in which positive charge would drift.
- V. Assume all batteries and meters are ideal unless otherwise stated.

AP[®] PHYSICS 1 EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_n|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$K = \frac{1}{2} m v^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$x = A \cos(2\pi f t)$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$L = I \omega$$

$$\Delta L = \tau \Delta t$$

$$K = \frac{1}{2} I \omega^2$$

$$|\vec{F}_s| = k |\vec{x}|$$

$$U_s = \frac{1}{2} k x^2$$

$$\rho = \frac{m}{V}$$

a = acceleration
 A = amplitude
 d = distance
 E = energy
 f = frequency
 F = force
 I = rotational inertia
 K = kinetic energy
 k = spring constant
 L = angular momentum
 ℓ = length
 m = mass
 P = power
 p = momentum
 r = radius or separation
 T = period
 t = time
 U = potential energy
 V = volume
 v = speed
 W = work done on a system
 x = position
 y = height
 α = angular acceleration
 μ = coefficient of friction
 θ = angle
 ρ = density
 τ = torque
 ω = angular speed

$$\Delta U_g = m g \Delta y$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$\vec{g} = \frac{\vec{F}_g}{m}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I \Delta V$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

A = area
 F = force
 I = current
 ℓ = length
 P = power
 q = charge
 R = resistance
 r = separation
 t = time
 V = electric potential
 ρ = resistivity

WAVES

$$\lambda = \frac{v}{f}$$

f = frequency
 v = speed
 λ = wavelength

GEOMETRY AND TRIGONOMETRY

Rectangle
 $A = bh$
 A = area
 C = circumference
 V = volume

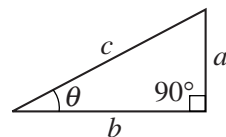
Triangle
 $A = \frac{1}{2} bh$
 S = surface area
 b = base
 h = height
 ℓ = length

Circle
 $A = \pi r^2$
 $C = 2\pi r$
 w = width
 r = radius

Rectangular solid
 $V = \ell wh$
Right triangle
 $c^2 = a^2 + b^2$

Cylinder
 $V = \pi r^2 \ell$
 $S = 2\pi r \ell + 2\pi r^2$
 $\sin \theta = \frac{a}{c}$
 $\cos \theta = \frac{b}{c}$

Sphere
 $V = \frac{4}{3} \pi r^3$
 $S = 4\pi r^2$
 $\tan \theta = \frac{a}{b}$



PHYSICS 1
Section I
40 Questions
Time—90 minutes

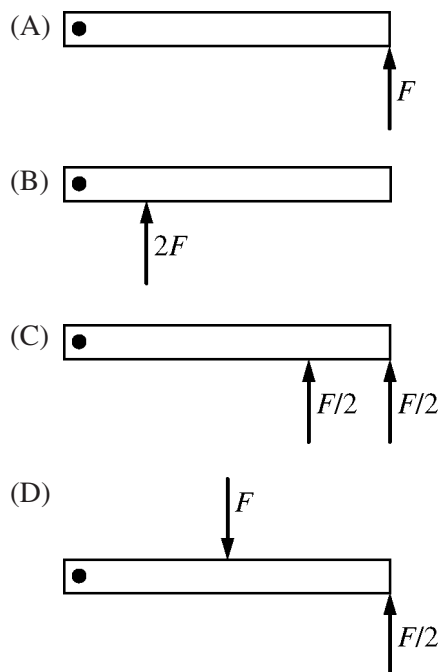
Note: To simplify calculations, you may use $g = 10 \text{ m/s}^2$ in all problems.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

1. A kitten sits in a lightweight basket near the edge of a table. A person accidentally knocks the basket off the table. As the kitten and basket fall, the kitten rolls, turns, kicks, and catches the basket in its claws. The basket lands on the floor with the kitten safely inside. If air resistance is negligible, what is the acceleration of the kitten-basket system while the kitten and basket are in midair?

- (A) The acceleration is directed downward with magnitude less than g because the basket is light.
(B) The acceleration is directed downward with magnitude equal to g because the system is a projectile.
(C) The acceleration fluctuates because of the rolling, turning, and kicking motion of the kitten.
(D) The acceleration cannot be determined without knowing how hard the basket is pushed.

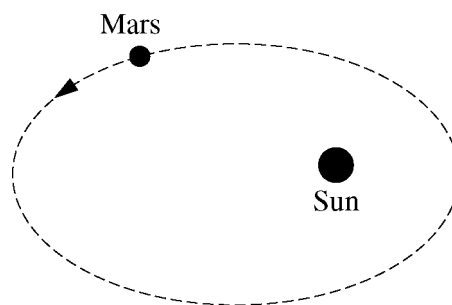
2. A solid metal bar is at rest on a horizontal frictionless surface. It is free to rotate about a vertical axis at the left end. The figures below show forces of different magnitudes that are exerted on the bar at different locations. In which case does the bar's angular speed about the axis increase at the fastest rate?



Speed	10 m/s	20 m/s	30 m/s
Braking Distance	6.1 m	23.9 m	53.5 m

3. To analyze the characteristics and performance of the brakes on a 1500 kg car, researchers collected the data shown in the table above. It shows the car's speed when the brakes are first applied and the corresponding braking distance required to stop the car. The magnitude of the average braking force on the car is most nearly

(A) 75,000 N
(B) 30,000 N
(C) 12,000 N
(D) 1600 N



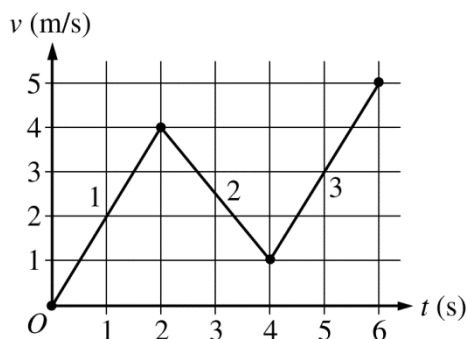
Note: Figure not drawn to scale.

4. Mars moves in an elliptical orbit around the Sun, and the mass of Mars is much less than the mass of the Sun. At the instant shown above, Mars is getting farther away from the Sun. How does this affect the potential energy of the Mars-Sun system and the magnitude of Mars's angular momentum with respect to the Sun?

System Potential <u>Energy</u>	Mars's Angular <u>Momentum</u>
--------------------------------------	--------------------------------------

- | | |
|---------------|------------------|
| (A) Increases | Increases |
| (B) Increases | Remains the same |
| (C) Decreases | Decreases |
| (D) Decreases | Remains the same |
5. An ice-skater is moving at a constant velocity across an icy pond. The skater throws a snowball directly ahead. Which of the following correctly describes the velocity of the center of mass of the skater-snowball system immediately after the snowball is thrown? Assume friction and air resistance are negligible.
- (A) It is equal to the velocity of the snowball.
(B) It is equal to the new velocity of the skater.
(C) It is equal to half the original velocity of the skater.
(D) It is equal to the original velocity of the skater.

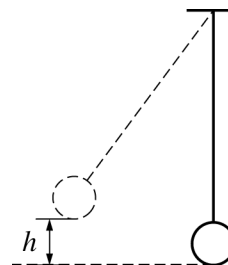
Questions 6-8 refer to the following material.



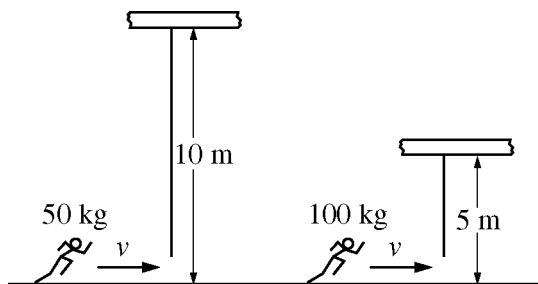
The graph above shows velocity v as a function of time t for a 0.50 kg object traveling along a straight line. The graph has three segments labeled 1, 2, and 3. A rope exerts a constant force of magnitude F_T on the object along its direction of motion the whole time. During segment 2 only, a frictional force of magnitude F_f is also exerted on the object.

6. Which of the following correctly ranks the displacement Δx for the three segments of the object's motion?
- (A) $\Delta x_3 > \Delta x_2 > \Delta x_1 > 0$
 (B) $\Delta x_1 = \Delta x_2 = \Delta x_3 > 0$
 (C) $(\Delta x_1 = \Delta x_3) > \Delta x_2 > 0$
 (D) $(\Delta x_1 = \Delta x_3) > 0 > \Delta x_2$
7. Which of the following expressions correctly relates the magnitudes F_f and F_T ?
- (A) $F_f < F_T$
 (B) $F_f = F_T$
 (C) $F_T < F_f < 2F_T$
 (D) $F_f = 2F_T$

8. For another identical object initially at rest, no frictional force is exerted during segment 2 (between $t = 2\text{ s}$ and $t = 4\text{ s}$). A rope exerts the same constant force of magnitude F_T as in the previous scenario. What is the change in the object's kinetic energy during segment 2?
- (A) 3.75 J
 (B) 4.00 J
 (C) 12.0 J
 (D) 16.0 J

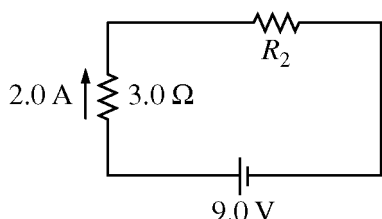


9. The pendulum shown in the figure above reaches a maximum height h above the equilibrium position as it oscillates. Assuming friction and air resistance are negligible, which of the following is true about the total energy of the Earth-pendulum system as the pendulum oscillates?
- (A) It is at a maximum when the pendulum is at its lowest position.
 (B) It is at a maximum when the pendulum is at its maximum height h .
 (C) It is constant throughout the pendulum's motion.
 (D) It is at a minimum when the pendulum is somewhere between its lowest and highest positions.



10. A 50 kg athlete running at speed v grabs a light rope that hangs from a 10-meter-high platform and swings to a maximum of 1.8 m above the ground. Later, a 100 kg athlete, running at the same speed, grabs a similar rope hanging from a 5-meter-high platform. What is the maximum height to which the 100 kg athlete swings?

(A) 0.9 m
(B) 1.8 m
(C) 2.5 m
(D) 3.6 m



11. A 9.0 V battery and two resistors are connected in the circuit shown above. The current through the $3.0\ \Omega$ resistor is 2.0 A. What is the potential difference across resistor R_2 ?

(A) 9.0 V
(B) 6.0 V
(C) 3.0 V
(D) 1.5 V

12. A student is asked to determine the work done on a block of wood when the block is pulled horizontally using an attached string. The student is supplied with a spring scale, a stopwatch, and a meterstick. Which of the following graphical analysis techniques will allow the student to determine the work done on the block by the string?

(A) Graphing the force as a function of time and calculating the slope
(B) Graphing the force as a function of time and calculating the area under the curve
(C) Graphing the force as a function of distance and calculating the slope
(D) Graphing the force as a function of distance and calculating the area under the curve

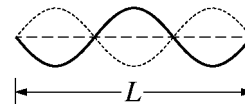
13. Block A and block B move toward each other on a level frictionless track. Block A has mass m and velocity $+v$. Block B has mass $2m$ and velocity $-v$. The blocks collide, and during the collision the magnitude of the net force exerted on block A is F . What is the magnitude of the net force exerted on block B , and why does it have that value?

(A) $2F$, because the mass of block B is twice that of block A and the blocks have the same acceleration during the collision.
(B) $F/2$, because the mass of block B is twice that of block A and the blocks have the same acceleration during the collision.
(C) F , because the blocks have the same speed immediately before the collision.
(D) F , because the net force is equal to the mutual contact force between the blocks.

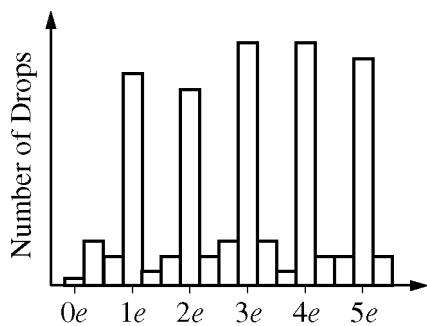
14. The data in the table below were recorded during an experiment in which two carts on a frictionless one-dimensional track collide head-on. What are the values of the magnitude of the change in momentum Δp_2 of cart 2 and the magnitude of its average acceleration a_2 during the collision?

	Cart 1	Cart 2
Mass	5 kg	1 kg
Average Force	15 N	15 N
Change in Momentum	0.3 kg•m/s	Δp_2
Average Acceleration	3 m/s ²	a_2

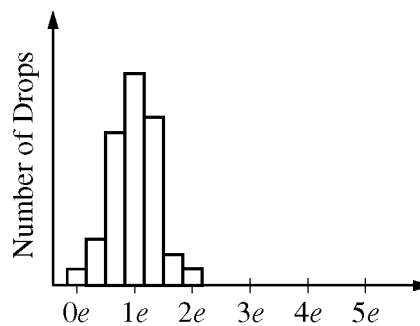
- | | |
|----------------|---------------------|
| Δp_2 | a_2 |
| (A) 0.3 kg•m/s | 3 m/s ² |
| (B) 0.3 kg•m/s | 15 m/s ² |
| (C) 1.5 kg•m/s | 3 m/s ² |
| (D) 1.5 kg•m/s | 15 m/s ² |
15. A 1.0 kg lump of clay is sliding to the right on a frictionless surface with speed 2 m/s. It collides head-on and sticks to a 0.5 kg metal sphere that is sliding to the left with speed 4 m/s. What is the kinetic energy of the combined objects after the collision?
- (A) 6 J
(B) 4 J
(C) 2 J
(D) 0 J



16. A guitar string of length L can vibrate with three antinodes, as shown above. The straight dashed line shows the equilibrium position of the string. The wave pattern is most likely formed by the superposition of which of the following pulses or waves?
- (A)
- (B)
- (C) Two periodic waves of wavelength $L/3$, one moving to the left and one moving to the right
- (D) Two periodic waves of wavelength $2L/3$, one moving to the left and one moving to the right
17. A speaker is at rest at the side of a straight road. The speaker produces a steady sound of constant frequency. A student is standing still near the speaker. Two vehicles are driven down the road: a truck approaching the speaker and a car moving away from the speaker. If the sound is represented by wave fronts, the student and the vehicle drivers each observe a different amount of time between successive wave fronts: t_s for the student, t_t for the truck driver, and t_c for the car driver. Which of the following correctly ranks the times?
- (A) $t_s < t_t < t_c$
(B) $t_c < t_s < t_t$
(C) $t_t < t_s < t_c$
(D) $t_t < t_c < t_s$



Graph 1



Graph 2

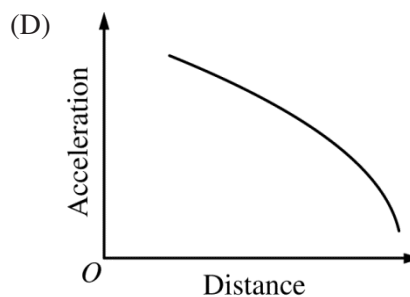
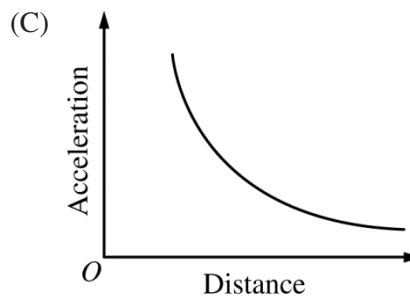
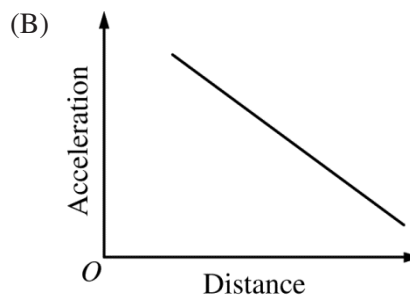
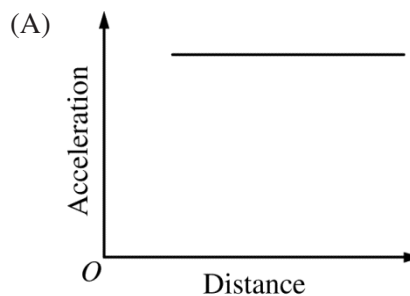
18. Tiny drops of oil that have different amounts of electric charge produced by a random process are sprayed into a region between a positively and a negatively charged plate. The charge on each drop is determined by analyzing the effect of the electric force on the drop's motion. Two graphs of the number of drops as a function of the charge on each drop, where e is the charge of an electron, are shown above. Which graph shows the most likely result after repeating the experiment many times, and why?
- (A) Graph 1, because it shows peaks with approximately the same amplitude
 - (B) Graph 1, because it shows that within experimental error charge only exists in multiples of e
 - (C) Graph 2, because it shows a peak value at the charge of an electron
 - (D) Graph 2, because it shows a narrower range of measurement error

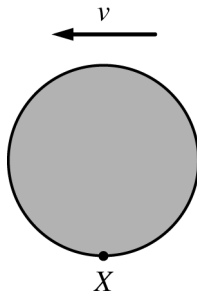
19. The magnitude of the gravitational field on the surface of a particular planet is $2g$. The planet's mass is half the mass of Earth. What is the planet's radius in terms of the radius R_E of Earth?

- (A) $R_E/4$
 (B) $R_E/2$
 (C) $R_E/\sqrt{2}$
 (D) $2R_E$



20. A small object with positive charge Q is fixed in place. A small bead with positive charge q is released from rest from the position shown above. In the absence of forces other than the electric force, which of the following graphs best represents the bead's acceleration as a function of its distance from the fixed object?



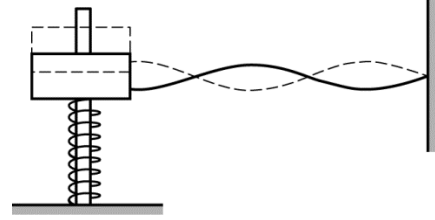


Top View

21. A stone disk is sliding on frictionless ice to the west with speed v , as shown in the figure above. As the disk slides by, a child uses a rubber mallet to hit the disk at point X , exerting a force directly toward the center of the disk. The child hits point X every half second for about 10 s, changing the trajectory of the disk but not causing it to rotate. Which of the following most closely approximates the path of the disk while the child is hitting it?

- (A) A northward linear path
- (B) A northwestward linear path
- (C) A parabolic path
- (D) A circular path

Questions 22-23 refer to the following material.

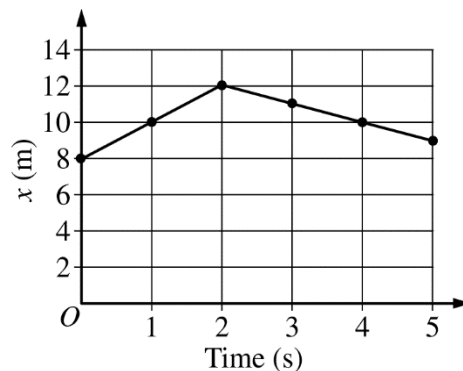


The figure above shows a pole with a spring around it and a 0.10 kg block with a hole in the middle on top of the spring. A light horizontal string is attached to the block and a wall. When the block is oscillating at 5.0 Hz, the standing wave shown is formed.

22. The spring constant of the spring is approximately which of the following?
- (A) 100 N/m
(B) 10 N/m
(C) 1.0 N/m
(D) 0.1 N/m
23. What additional measurement is needed to determine the speed of the wave on the string?
- (A) The mass of the string
(B) The effective spring constant of the spring
(C) The distance between the wall and the block
(D) The amplitude of the block's oscillation

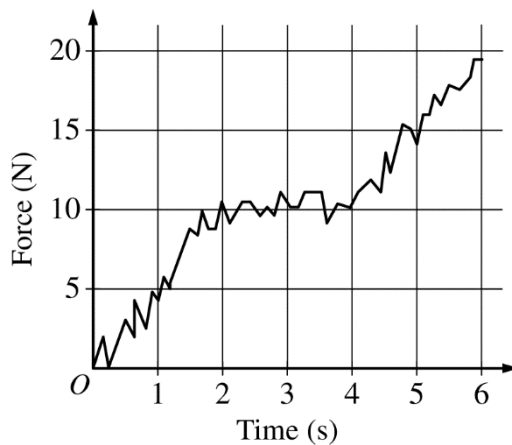
24. Two identical blocks with mass 5.0 kg each are connected to the opposite ends of a compressed spring. The blocks initially slide together on a frictionless surface with velocity 2 m/s to the right. The spring is then released by remote control. At some later instant, the left block is moving at 1 m/s to the left, and the other block is moving to the right. What is the speed of the center of mass of the system at that instant?

- (A) 4 m/s
- (B) 3 m/s
- (C) 2 m/s
- (D) 0 m/s

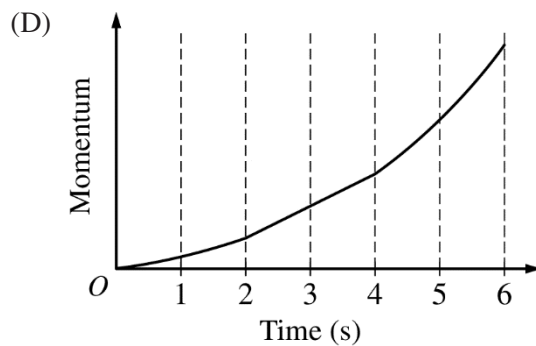
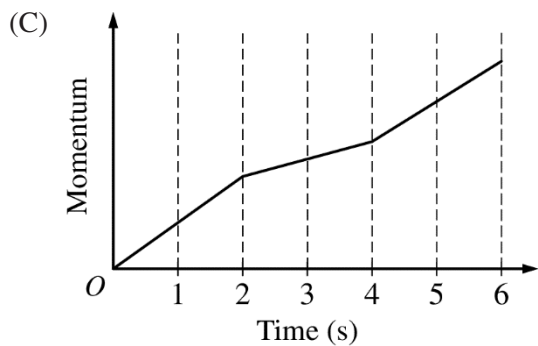
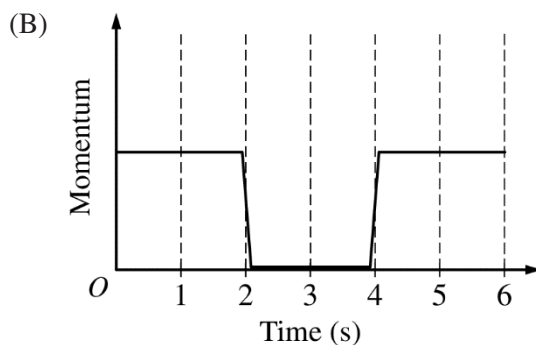
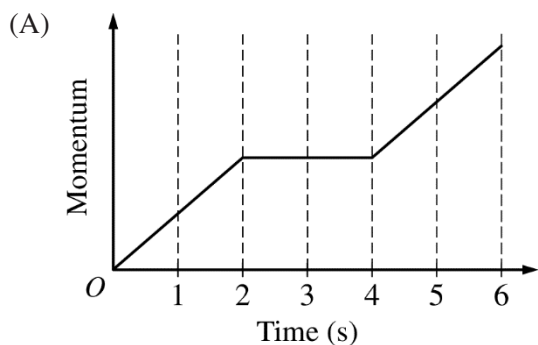


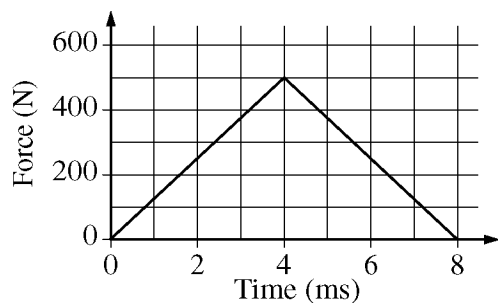
25. The graph above shows the position x as a function of time for the center of mass of a system of particles of total mass 6.0 kg . For a very short time interval around 2.0 s , an external force is exerted on an object in the system. What is the resulting change in momentum of the system?

- (A) $52\text{ kg}\cdot\text{m/s}$
- (B) $6\text{ kg}\cdot\text{m/s}$
- (C) $-6\text{ kg}\cdot\text{m/s}$
- (D) $-18\text{ kg}\cdot\text{m/s}$



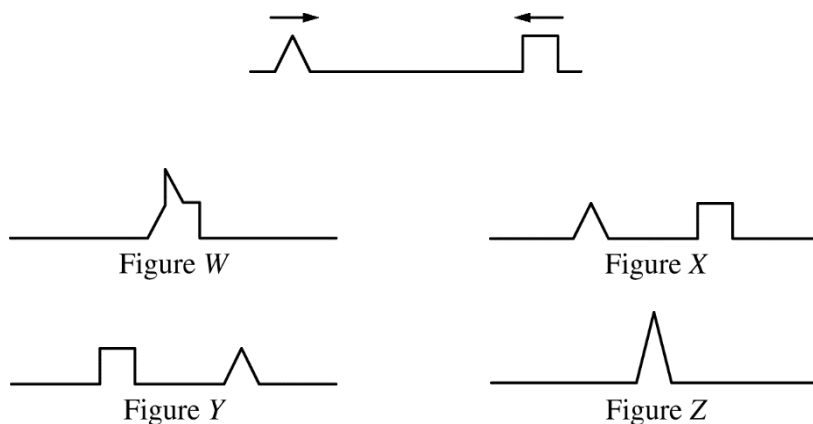
26. Using a force probe, a student generates the graph above of the force exerted on a small wagon as a function of time. The wagon starts from rest and rolls with negligible friction in the axles. Which of the following graphs best represents the wagon's momentum as a function of time?





27. A 0.050 kg tennis ball is moving to the left at 10 m/s when it is hit by a tennis racket that is moving to the right. The magnitude of the force exerted on the ball by the racket as a function of time is shown in the figure above. What is the speed of the ball after the collision with the racket?

(A) 30 m/s
(B) 50 m/s
(C) 70 m/s
(D) 90 m/s

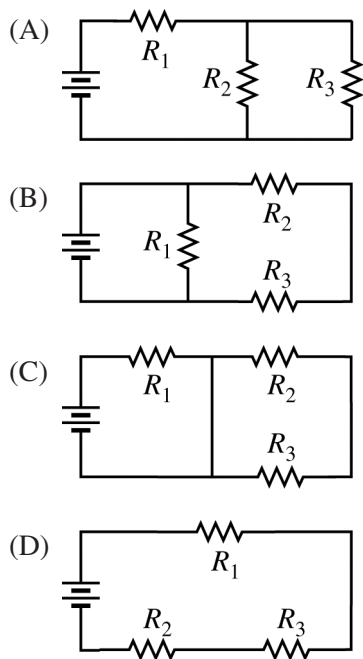


28. Two wave pulses are created simultaneously at opposite ends of a string and move toward the center of the string, as shown in the top figure above. Both pulses have the same width. The labeled figures show possible shapes of the string at different moments in time. Which of the following indicates a correct time order for the shape of the string as the pulses move along the string, come together, and move apart?

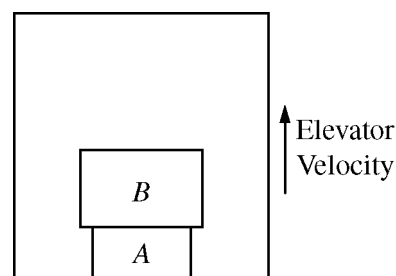
- (A) Figure X, Figure W, Figure X
- (B) Figure X, Figure W, Figure Y
- (C) Figure X, Figure Z, Figure X
- (D) Figure X, Figure Z, Figure Y

Circuit Element	Potential Difference (V)	Current (A)
Battery	10	8
Resistor R_1	6	8
Resistor R_2	4	3
Resistor R_3	4	5

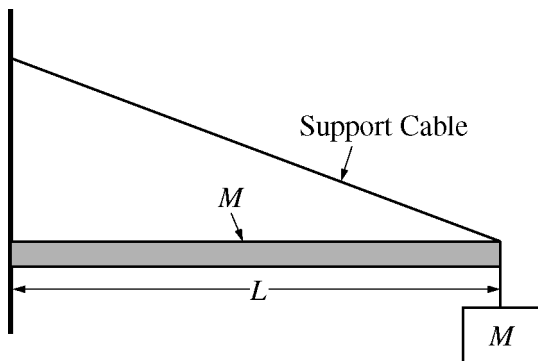
29. The above table shows the potential difference and current for each circuit element in a circuit consisting of a battery and three resistors. Which of the following diagrams could represent the circuit?



30. An object starts from rest and slides with negligible friction down an air track tipped at an angle θ from the horizontal. A student records values of the object's position along the track at various times. The value of θ can best be determined from which of the following?
- (A) The y-intercept of a graph of position as a function of time
- (B) The y-intercept of a graph of position as a function of the square of time
- (C) The slope of a graph of position as a function of time
- (D) The slope of a graph of position as a function of the square of time



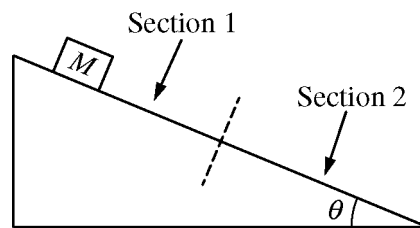
31. Box A of mass m sits on the floor of an elevator, with box B of mass $2m$ on top of it, as shown in the figure above. The elevator is moving upward and slowing down. F_A is the magnitude of the force exerted on box A by box B, F_B is the magnitude of the force exerted on box B by box A, and F_g is the magnitude of the gravitational force exerted on box B. Which of the following ranks the forces in order of increasing magnitude?
- (A) $F_B = F_A = F_g$
- (B) $(F_B = F_A) < F_g$
- (C) $F_B < (F_A = F_g)$
- (D) $F_g < F_B < F_A$



32. The figure above shows a uniform beam of length L and mass M that hangs horizontally and is attached to a vertical wall. A block of mass M is suspended from the far end of the beam by a cable. A support cable runs from the wall to the outer edge of the beam. Both cables are of negligible mass. The wall exerts a force F_W on the left end of the beam. For which of the following actions is the magnitude of the vertical component of F_W smallest?

- (A) Keeping the support cable and block as shown in the diagram
- (B) Moving the lower end of the support cable to the center of the beam and leaving the block at the outer end of the beam
- (C) Keeping the lower end of the support cable at the outer end of the beam and moving the block to the center of the beam
- (D) Moving both the support cable and the block to the center of the beam

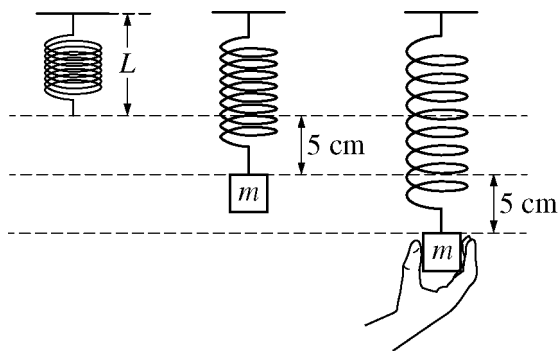
Questions 33-34 refer to the following material.



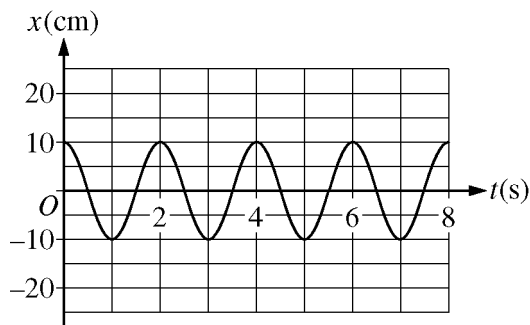
The inclined plane in the figure above has two sections of equal length and different roughness. The dashed line shows where section 1 ends and section 2 begins. A block of mass M is placed at different locations on the incline. The coefficients of kinetic and static friction between the block and each section are shown in the table below.

Coefficient of Friction	Section 1	Section 2
Static	μ_{S1}	$\mu_{S2} (> \mu_{S1})$
Kinetic	μ_k	$2\mu_k$

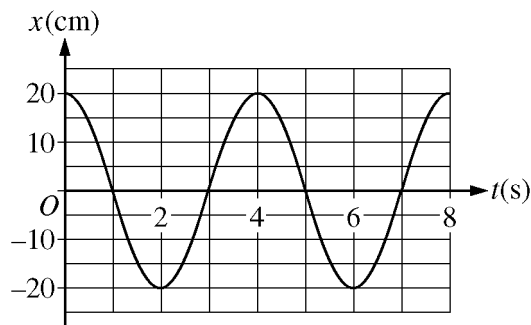
33. If the block is at rest on section 1 of the incline, what is the magnitude of the force of static friction exerted on the block by the incline?
- (A) $\mu_{S1}Mg \cos \theta$
 - (B) $\mu_{S1}Mg \tan \theta$
 - (C) $Mg \sin \theta$
 - (D) $Mg / \tan \theta$
34. If the block is sliding up section 2, what is the magnitude of the force of friction that is exerted on the block by the incline?
- (A) $2\mu_kMg \cos \theta$
 - (B) $2\mu_kMg \tan \theta$
 - (C) $Mg \sin \theta$
 - (D) $Mg / \tan \theta$



35. An object with mass m is suspended at rest from a spring with a spring constant of 200 N/m . The length of the spring is 5.0 cm longer than its unstretched length L , as shown above. A person then exerts a force on the object and stretches the spring an additional 5.0 cm . What is the total energy stored in the spring at the new stretched length?
- (A) 0.25 J
(B) 1.0 J
(C) 10 J
(D) 20 J



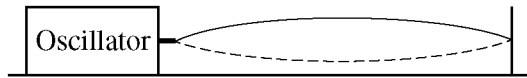
Block A



Block B

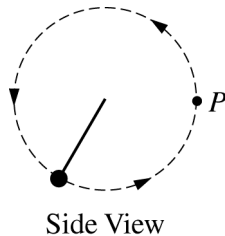
36. Two blocks are connected to identical ideal springs and are oscillating on a horizontal frictionless surface. Block A has mass m , and its motion is represented by the graph of position as a function of time shown above on the left. Block B's motion is represented above on the right. Which of the following statements comparing block B to block A is correct?
- (A) Because block B covers more distance per cycle than block A, block B takes more time to complete each cycle.
 - (B) Because the spring attached to block B is initially stretched a greater distance, the spring constant is smaller and therefore block B has a slower average speed than block A does.
 - (C) Because block B has more mass, it has a slower average speed than block A does.
 - (D) Because block B has more mass, its acceleration is smaller than that of block A at any given displacement from the equilibrium position.

Directions: For each of the questions or incomplete statements below, two of the suggested answers will be correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on page 3 of the answer sheet.



131. The figure above shows a string held taut between a post and an oscillator that vibrates the end of the string. In an experiment, the oscillator is adjusted to find many frequencies that will create a standing wave, one of which is shown. The number of nodes in each standing wave is counted. Which of the following quantities must also be measured to determine the speed of traveling waves on the string? Select two answers.

- (A) The length of the string
- (B) The mass of the string
- (C) The amplitudes of the standing wave patterns
- (D) The frequencies of the standing wave patterns

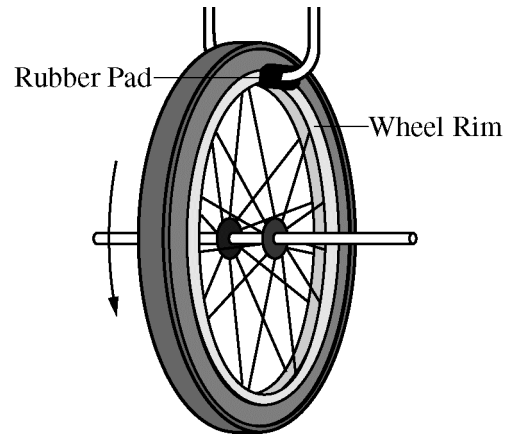


132. A ball attached to a light string swings in a counterclockwise vertical circle, as shown above. Which of the following arrows represent one of the forces exerted on the ball at the moment it passes through point P ? Select two answers.

- (A)
- (B)
- (C)
- (D)

133. Object A has mass 2 kg and is traveling to the right at speed 3 m/s . Object B is traveling to the left at 3 m/s . The objects collide head-on, and afterward each has a speed of 3 m/s . Which of the following could be the mass of object B ? Select two answers.

- (A) 3 kg
- (B) 2 kg
- (C) 1 kg
- (D) Negligible compared to 2 kg

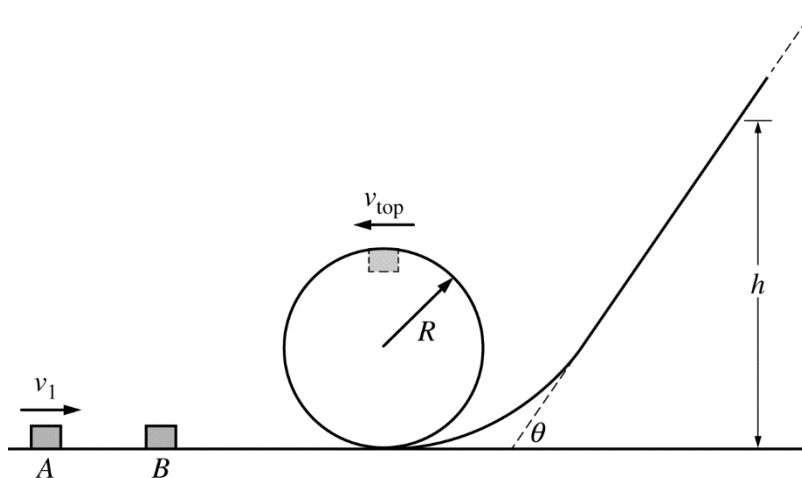


134. Some bicycle brakes work by pressing rubber pads against the rim of the wheel. To test newly designed brakes, bicycle engineers mount a wheel of known rotational inertia on a low-friction axle, as shown above. The engineers spin the wheel with known initial angular speed and then apply the brakes with a constant force. Which of the following procedures would enable the engineers to find the torque exerted by the brakes on the wheel? Select two answers.

- (A) Measuring how much time the wheel takes to come to rest
- (B) Measuring how many rotations the wheel completes while coming to rest
- (C) Measuring the distance from axle to brakes and the normal force between the rubber pads and the rim
- (D) Measuring the mechanical energy dissipated as the rim rubs against the rubber pads

PHYSICS 1
Section II
5 Questions
Time—90 minutes

Directions: Questions 1, 4 and 5 are short free-response questions that require about 13 minutes each to answer and are worth 7 points each. Questions 2 and 3 are long free-response questions that require about 25 minutes each to answer and are worth 12 points each. Show your work for each part in the space provided after that part.



Note: Figure not drawn to scale.

1. (7 points, suggested time 13 minutes)

Two blocks A and B of equal mass m are on a frictionless track, as shown in the figure above. Block A , initially moving with speed v_1 , has a perfectly elastic collision with block B . Block B has a speed v_1 immediately after the collision, and then it travels around a circular loop of radius R , where R is much larger than the size of the blocks. The speed of block B at the top of the loop is v_{top} . Block B then slides up a ramp until it comes momentarily to rest at a height h above the floor.

- (a) Derive an equation for the height h in terms of m , R , v_{top} , and g , as appropriate.

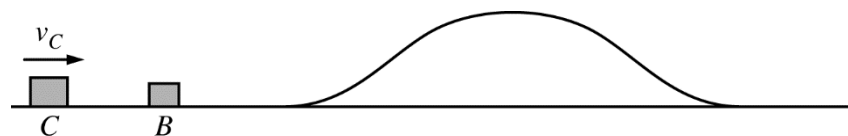
- (b) Suppose that the speed v_1 of block A is doubled. Will the final height of block B be greater than, less than, or equal to $2h$?

_____ Greater than $2h$ _____ Less than $2h$ _____ Equal to $2h$

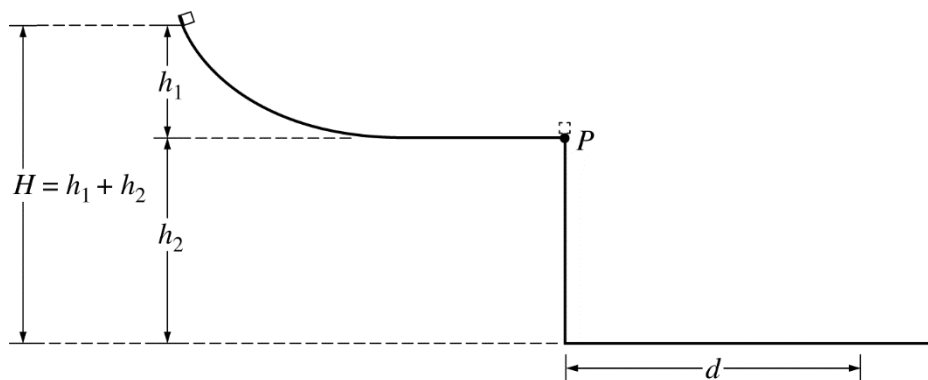
Briefly explain how you arrived at your answer.



- (c) Blocks A and B (both of mass m) are moved to a different frictionless track, as shown above. Let v_A be the minimum initial speed for block A that allows block B to make it over the hump in the track. Block A is then replaced with a larger block, block C , which has mass $2m$, as shown below.



Block C makes a completely inelastic collision with block B so that both blocks stick together and travel along the track. What is the minimum initial speed v_C that block C must have so that the two-block system makes it over the hump? Express your answer in terms of v_A .



2. (12 points, suggested time 25 minutes)

A student releases a block of mass m from rest at the top of a slide of height h_1 . The block moves down the slide and off the end of a table of height h_2 , landing on the floor a horizontal distance d from the edge of the table. Friction and air resistance are negligible. The overall height H of the setup is determined by the height of the room. Therefore, if h_1 is increased, h_2 must decrease by the same amount so that the sum $h_1 + h_2$ remains equal to H . The student wants to adjust h_1 and h_2 to make d as large as possible.

(a)

i. Without using equations, explain why making h_1 very small would cause d to be small, even though h_2 would be large.

ii. Without using equations, explain why making h_2 very small would cause d to be small, even though h_1 would be large.

(b) Derive an equation for d in terms of h_1 , h_2 , m , and physical constants, as appropriate.

(c)

- i. Write the equation or step in your derivation in part (b) (not your final answer) that supports your reasoning in part (a)i.

Briefly explain your choice.

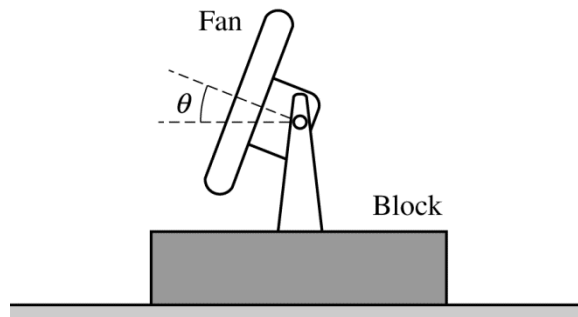
- ii. Write the equation or step in your derivation in part (b) (not your final answer) that supports your reasoning in part (a)ii.

Briefly explain your choice.

- (d) If the experiment is repeated on the Moon without changing h_1 or h_2 , will the new landing distance d be greater than, less than, or the same as the landing distance when the experiment is performed on Earth?

_____ Greater than _____ Less than _____ The same as

Briefly explain how you arrived at your answer.

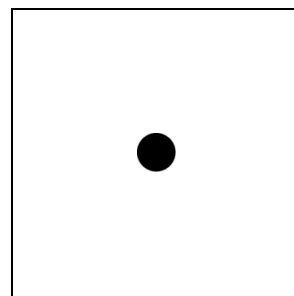


3. (12 points, suggested time 25 minutes)

A student in a physics lab has a block with a fan attached to it, as shown in the figure above. The fan has a pivot so that the angle θ it makes with the horizontal can be adjusted between 0° and 90° . When the fan is pointed horizontally so that $\theta = 0^\circ$, the block accelerates from rest along a track, even though there is friction between the block and the track. For the following questions, assume that the student has access to equipment that would usually be found in a school physics laboratory.

- (a) Describe an experimental procedure that the student could use to measure the force F that the air exerts on the fan-block system when the fan is turned on. Assume that the magnitude of this force is the same for all angles θ .
 - i. What quantities would be measured?
 - ii. What equipment would be used for the measurements, and how would that equipment be used? Include a labeled diagram of the experimental setup.
 - iii. Describe the overall procedure to be used. Give enough detail so that another student could replicate the experiment.
 - iv. Describe how the force can be determined from the measurements described in parts (a)i–iii.

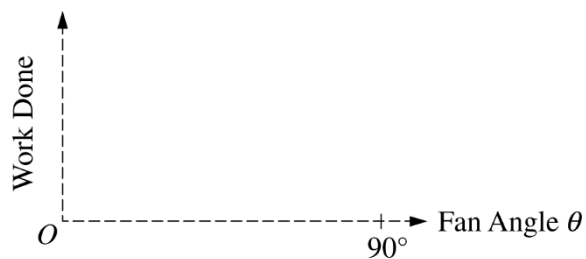
- (b) Describe an experimental procedure that would allow the student to use the fan-block system, with the fan turned on, to find the coefficient of kinetic friction between the block and the track.
- What quantities would be measured?
 - What equipment would be used for the measurements, and how would that equipment be used? Include a labeled diagram of the experimental setup.
 - Describe the overall procedure to be used. Give enough detail so that another student could replicate the experiment.
 - On the dot to the right, which represents the fan-block system, draw and label the forces (not components) that are exerted on the system during the experiment described in parts (b)i–iii. Represent each force by a distinct arrow starting on, and pointing away from, the dot.

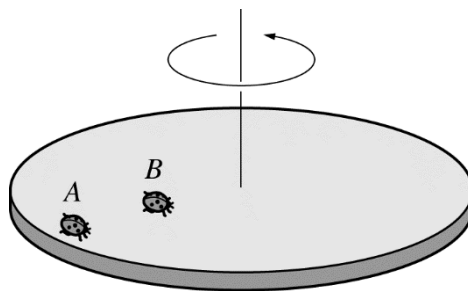


- (c) Describe how to analyze the data from the experiment described in parts (b)i–iii in order to determine the coefficient of kinetic friction between the block and the track. Your analysis may include the force F exerted by the air on the fan-block system. Do not add to the free-body diagram in part (b)iv.

Question 3 continues on next page.

- (d) In a different experiment, the student attaches the block to the top of a low-friction cart. On the axes below, sketch a graph of the work done by air on the fan-block-cart system as a function of θ when the system travels a fixed distance D .





4. (7 points, suggested time 13 minutes)

Two ladybugs are standing on a rotating disk that is spinning counterclockwise, as shown in the figure above. Assume that friction in the bearings of the axle is negligible.

(a)

- i. Is the angular speed of ladybug *A* greater than, less than, or the same as the angular speed of ladybug *B* ?

_____ Greater _____ Less _____ The same

Briefly justify your answer.

- ii. Is the linear speed of ladybug *A* greater than, less than, or the same as the linear speed of ladybug *B* ?

_____ Greater _____ Less _____ The same

Briefly justify your answer.

- (b) Ladybug A begins walking in a circular path in the direction of the disk's rotation. Does the magnitude of the angular momentum of the disk alone (not the ladybugs-disk system) increase, decrease, or stay the same?
- ____ Increase ____ Decrease ____ Stay the same
- Briefly explain your reasoning.

- (c) In a different scenario, a single ladybug is standing near the edge of the disk at a distance of $0.9R$ from the center, where R is the radius of the disk, as shown in Figure 1 below. The rotational inertia of the ladybug-disk system is I_1 , and the disk completes one rotation in 2.5 s. The ladybug then walks toward the center of the disk to a distance of $0.1R$ from the center and comes to a stop relative to the disk, as shown in Figure 2. Now the rotational inertia of the system is I_2 , and the disk completes one rotation every 2.0 s.

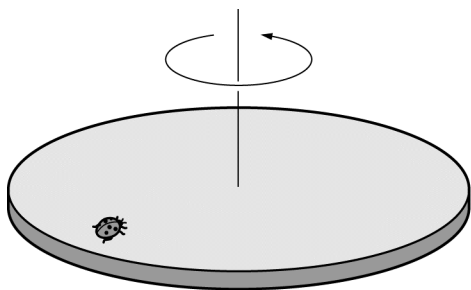


Figure 1

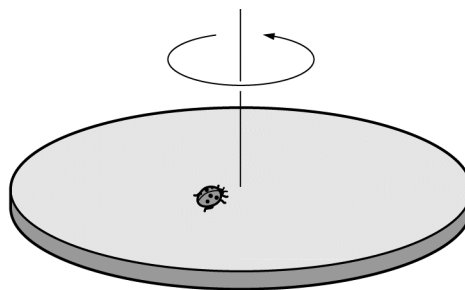


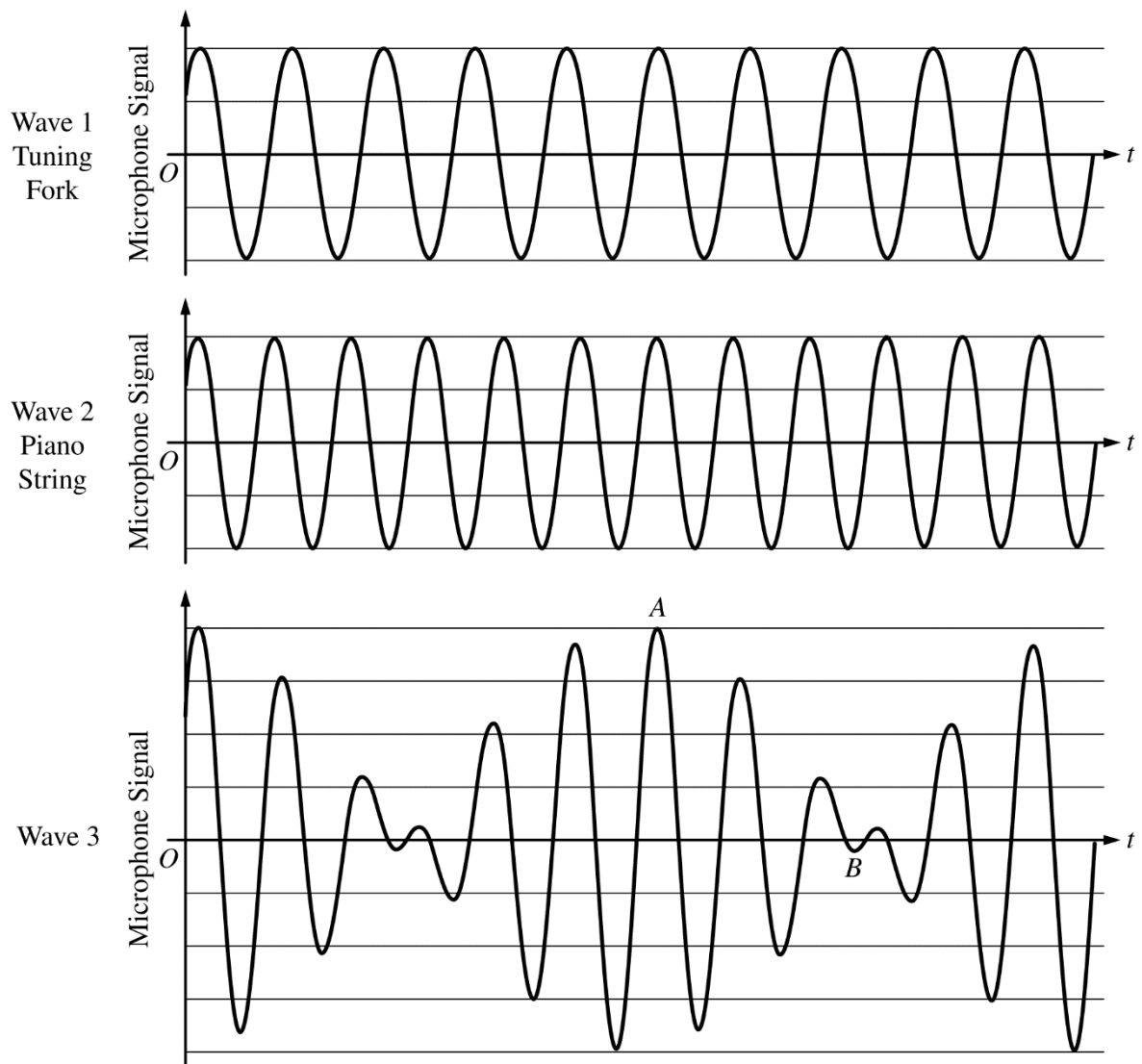
Figure 2

- i. Derive an equation for I_2 in terms of I_1 .

- ii. While the ladybug is walking toward the center of the disk, does it exert a torque on the disk?

____ Yes ____ No

Briefly explain your reasoning.



5. (7 points, suggested time 13 minutes)

The graphs shown above represent the variations in air pressure in the sound waves recorded by a microphone as functions of time t . Wave 1 represents the sound wave when a tuning fork has been struck, wave 2 represents the sound wave when a particular piano string has been struck, and wave 3 represents the sound wave when both the tuning fork and the piano string have been struck.

(a) Briefly explain why the peaks in wave 3 at time A and nearby times are relatively large.

- (b) Briefly explain why the peaks in wave 3 at time B and nearby times are relatively small.

If the fundamental frequency of the piano string is close to the frequency of the tuning fork, the phenomenon illustrated in wave 3 is detectable by ear. A piano tuner can loosen or tighten a piano string to change the fundamental frequency of the string.

- (c) In a clear, coherent paragraph-length response referring to the graphs on the previous page, describe how and why the graph of wave 3, including the locations of points A and B , would change as the piano tuner loosens or tightens the string. Your response should also describe how the sound heard by the piano tuner changes during the process of adjusting the frequency of the string and how the piano tuner can tell when the string's frequency matches that of the tuning fork.

Answer Key for AP Physics 1 Practice Exam, Section I

Question 1: B	Question 21: C
Question 2: A	Question 22: A
Question 3: C	Question 23: C
Question 4: B	Question 24: C
Question 5: D	Question 25: D
Question 6: A	Question 26: D
Question 7: C	Question 27: A
Question 8: C	Question 28: B
Question 9: C	Question 29: A
Question 10: B	Question 30: D
Question 11: C	Question 31: B
Question 12: D	Question 32: D
Question 13: D	Question 33: C
Question 14: B	Question 34: A
Question 15: D	Question 35: B
Question 16: D	Question 36: D
Question 17: C	Question 131: A, D
Question 18: B	Question 132: C, D
Question 19: B	Question 133: B, D
Question 20: C	Question 134: A, B

2016 AP Physics 1: Algebra-Based

Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
1	4.A.1.1; 4.A.2.1; 5.D.3.1	4.A.1; 4.A.2; 5.D.3	1.2; 1.4; 6.4; 6.4	B	80
2	3.F.2.1	3.F.2	6.4	A	67
3	3.A.1.1; 3.B.1.1; 5.B.4.1; 5.B.5.4	3.A.1; 3.B.1; 5.B.4; 5.B.5	1.5; 2.2; 6.4; 7.2; 6.4; 7.2; 6.4; 7.2	C	54
4	5.B.3.1; 5.E.1.1	5.B.3; 5.E.1	6.4; 7.2; 6.4	B	39
5	5.D.3.1	5.D.3	6.4	D	46
6	3.A.1.1; 4.A.2.3	3.A.1; 4.A.2	1.5; 2.2; 1.4; 2.2	A	38
7	3.A.3.1; 3.B.1.1; 4.A.2.1	3.A.3; 3.B.1; 4.A.2	6.4; 7.2; 6.4; 7.2; 6.4	C	37
8	5.B.4.2	5.B.4	1.4; 2.1; 2.2	C	30
9	5.B.4.1	5.B.4	6.4	C	67
10	5.B.4.2	5.B.4	1.4; 2.2	B	50
11	5.B.9.3	5.B.9	2.2	C	56
12	5.B.5.2	5.B.5	5.1	D	49
13	3.A.4.2	3.A.4	6.4; 7.2	D	47
14	3.A.1.1; 3.B.1.1; 3.D.2.1; 3.D.2.2; 5.D.1.4	3.A.1; 3.B.1; 3.D.1; 3.D.2; 5.D.1	2.2; 6.4; 2.1; 6.4; 5.1; 6.4	B	38
15	5.D.2.5	5.D.2	2.2	D	36
16	6.D.3.2	6.D.3	6.4	D	33
17	6.B.5.1	6.B.5	1.4	C	33
18	1.B.3.1	1.B.3	6.1	B	36
19	2.B.2.2	2.B.2	2.2	B	36
20	3.A.1.1; 3.B.1.1; 3.C.2.1	3.A.1; 3.B.1; 3.C.2	1.5; 2.2; 6.4; 7.2; 2.2; 6.4	C	44
21	3.A.1.1; 3.D.2.2	3.A.1; 3.D.2	1.5; 6.4	C	37
22	3.B.3.3	3.B.3	2.2	A	22
23	6.D.3.2; 6.D.4.2	6.D.3; 6.D.4	6.4; 2.2	C	62
24	5.D.3.1	5.D.3	6.4	C	40
25	3.A.1.1; 4.B.1.1	3.A.1; 4.B.1	1.5; 2.2; 2.2	D	28
26	3.A.1.1; 4.B.2.2	3.A.1; 4.B.2	1.5; 2.2; 5.1	D	25
27	4.B.1.1; 4.B.2.1	4.B.1; 4.B.2	1.4; 2.2; 2.2	A	26
28	6.D.1.1; 6.D.2.1	6.D.1; 6.D.2	1.4; 5.1	B	32
29	5.B.9.3; 5.C.3.3	5.B.9; 5.C.3	2.2; 6.4; 7.2; 1.4; 2.2	A	65
30	3.A.1.1; 3.A.1.3; 3.A.3.1; 3.B.2.1	3.A.1; 3.A.3; 3.B.2	1.5; 5.1; 6.4; 1.1; 1.4	D	23
31	3.A.4.2; 3.B.1.3	3.A.4; 3.B.1	6.4; 7.2; 1.5; 2.2	B	41
32	3.F.1.1	3.F.1	1.4	D	24
33	3.A.3.1; 3.B.2.1	3.A.3; 3.B.2	6.4; 1.1; 1.4; 2.2	C	24
34	3.A.3.1; 3.B.2.1; 3.B.3.1	3.A.3; 3.B.2; 3.B.3	6.4; 1.1; 1.4; 2.2; 6.4	A	55
35	5.B.3.2	5.B.3	1.4; 2.2	B	32
36	3.B.3.1	3.B.3	6.4	D	17

2016 AP Physics 1: Algebra-Based Question Descriptors and Performance Data

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
131	6.B.4.1; 6.D.3.3	6.B.4; 6.D.3	4.2; 5.1; 7.2; 5.1	A, D	57
132	3.A.2.1; 3.B.2.1	3.A.2; 3.B.2	1.1; 1.1; 1.4	C, D	35
133	5.D.1.5	5.D.1	2.2	B, D	48
134	4.D.2.2	4.D.2	4.2	A, B	27

Free-Response Questions

Question	Learning objectives	Essential Knowledge	Science Practices	Mean Score
1	5.B.3.1; 5.B.3.3; 5.B.4.1; 5.B.4.2; 5.D.2.3; 5.D.2.5	5.B.3; 5.B.4; 5.D.2	2.2; 6.4; 1.4; 2.2; 6.4; 1.4; 2.1; 2.2; 6.4; 7.2; 2.1; 2.2	1.63
2	3.A.1.1; 5.B.3.1; 5.B.3.3; 5.B.4.2; 5.B.5.5	1.A.1; 2.A.1; 3.A.1; 5.B.3; 5.B.4; 5.B.5	1.5; 2.1; 2.2; 2.2; 6.4; 7.2; 1.4; 2.2; 1.4; 2.2; 2.2; 6.4	2.76
3	3.A.1.2; 3.A.1.3; 3.A.2.1; 3.B.1.2; 3.B.2.1; 4.A.2.3; 4.A.3.1; 4.C.2.1; 4.C.2.2; 5.B.5.3	3.A.1; 3.A.2; 3.B.1; 3.B.2; 4.A.2; 4.A.3; 4.C.2; 5.B.5	4.2; 5.1; 1.1; 4.2; 5.1; 1.1; 1.4; 1.4; 2.2; 2.2; 6.4; 1.4; 2.2; 7.2; 1.4; 2.2; 6.4	3.96
4	3.A.1.1; 4.D.2.1; 4.D.3.1; 5.E.1.1; 5.E.1.2	3.A.1; 4.D.2; 4.D.3; 5.E.1; 5.E.2	1.5; 2.1; 2.2; 1.2; 1.4; 2.2; 6.4; 2.2	1.44
5	6.D.2.1; 6.D.5.1	6.D.2; 6.D.5	5.1; 1.2	2.28

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Question 1

7 points total

**Distribution
of points**

(a) 3 points

For using an equation expressing the conservation of energy

1 point

$$mgh = 2mgR + \frac{1}{2}mv_{\text{top}}^2 \text{ or } mgh = (1/2)mv_1^2 \text{ or } mgh = (1/2)mv^2$$

For using two forms of energy when block B is at the top of the loop or using the initial kinetic energy

1 point

For expressing the correct answer in terms of the listed quantities

1 point

$$h = \frac{2mgR + \frac{1}{2}mv_{\text{top}}^2}{mg}$$

$$h = 2R + \frac{v_{\text{top}}^2}{2g}$$

(b) 2 points

For a correct answer “Greater than $2h$ ” with an explanation

1 point

No point is earned for a correct selection without an explanation.

If an incorrect answer is selected, the following point for using conservation of energy can still be earned.

For using an equation or semi-quantitative reasoning expressing the conservation of energy Note: “semi-quantitative reasoning” here includes reasoning in terms of proportionality, squared relationships, etc.

1 point

$$mgh = \frac{1}{2}mv_1^2$$

$$h = \frac{v_1^2}{2g}$$

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Question 1 (continued)

**Distribution
of points**

(c) 2 points

Correct answer: $v_C = \frac{3}{2} v_A$

Note: there is no “answer point”, so no credit is awarded for an answer with no work shown

For using momentum conservation to relate the speeds before and after the completely inelastic collision between blocks *B* and *C* 1 point

$2mv_C = (2m + m)v_{BC} = 3mv_{BC}$, where v_{BC} is the speed of blocks *B* and *C* after the collision

$$2v_C = 3v_{BC}$$

For setting the speed of blocks *B* and *C* after the collision equal to the speed of block *B* after its elastic collision with block *A*, in order for blocks *B* and *C* to make it over the hump. 1 point

$$2v_C = 3v_A$$

$$v_C = \frac{3}{2} v_A$$

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Question 2

12 points total

**Distribution
of points**

(a)

i. 2 points

For recognizing a small horizontal velocity for the block at point P 1 point

For stating or implying that the block doesn't go very far even though there is a longer fall time 1 point

Example: Even though the large h_2 allows the block to stay in the air for a long time, the small h_1 means the system loses little gravitational potential energy and hence gains little kinetic energy so that it leaves the table with too little speed to cover much distance while aloft.

ii. 2 points

For recognizing a large horizontal velocity for the block at point P 1 point

For mentioning a short time of flight for the falling block 1 point

Example: Now the block gains a lot of kinetic energy on the ramp so that it leaves the table with a large horizontal speed. But the small h_2 means the block spends so little time in the air that it lands before covering much horizontal distance.

(b) 3 points

For using an energy conservation statement (corresponds to step 1 in example below) 1 point

For setting up a time of flight calculation using vertical acceleration (corresponds to step 2a in example below) 1 point

Note: derivation of t must be shown to earn credit.

For using $vt = d$ to determine distance traveled (corresponds to step 2b in example below) 1 point

Example:

Step 1: Use energy conservation to find the block's speed at the bottom of ramp, which equals the launch speed v_1 at point P .

$$mgh_1 = \frac{1}{2}mv_1^2 \Rightarrow v_1 = \sqrt{2gh_1}$$

Step 2a: Find the time of flight, using the independence of the horizontal and vertical motion. Since the block leaves the table with zero *vertical* speed, the vertical

kinematic equation $y = y_0 + v_{y0}t + \frac{1}{2}a_yt^2$ reduces to $h_2 = \frac{1}{2}gt^2$ (taking

downward as positive). Solve for time to get $t = \sqrt{\frac{2h_2}{g}}$.

Step 2b. Find the horizontal distance d covered while in the air after point P . Since there are no horizontal forces exerted on the block, its horizontal velocity stays constant at v_1 . So, the block travels a horizontal distance

$$d = v_1t = \sqrt{2gh_1}\sqrt{\frac{2h_2}{g}} = 2\sqrt{h_1h_2}$$

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Question 2 (continued)

**Distribution
of points**

(c)

i. 2 points

For mentioning a correct equation or step of reasoning (energy conservation on ramp)	1 point
For correctly explaining why this equation or step mirrors the reasoning of (a)i	1 point
Example: My energy conservation reasoning (part (b) step 1) mirrors my reasoning that a small ramp height corresponds to a low potential energy (mgh_1), which is converted into low kinetic energy and hence a low speed.	

ii. 2 points

For mentioning a correct equation or step of reasoning (time of flight)	1 point
For correctly explaining why that equation or step mirrors the reasoning of (a)ii	1 point
Example: My time-of-flight calculation (part (b) step 2a) shows that t is proportional to $\sqrt{h_2}$, so a smaller falling distance h_2 corresponds to a smaller flight time.	

(d) 1 point

Correct answer: “The same as”	
Note: explanation is scored regardless of checkbox.	
Point can be earned if response is consistent with answer in part (b)	
For reasoning that is consistent with the functional dependence of the part (b) answer	1 point

Example:

As found in part (b), the distance d does not depend on g . So, doing the experiment on the Moon instead of Earth makes no difference. The increased time of flight (since weaker gravity makes the block take longer to land) compensates for the small speed gained by the block on the ramp.

Checked “Greater than” then g must be in the denominator in the incorrect equation in part (b)

Checked “Less than” then g must be in the numerator in the incorrect equation in part (b)

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Question 3

12 points total

**Distribution
of points**

(a)

i, ii, and iii) 2 points

Note: Parts i, ii, and iii are read together

For measuring the mass or weight of the relevant system

1 point

For making a measurement of force (or something from which a force can be calculated)
with the fan turned on

1 point

Note: fan angles of either 0° or 90° need not be measured directly, since these are given as the minimum and maximum angles possible. If the procedure involves intermediate angles, then these angles must be measured.

Example 1 (variant - Example 1a in parentheses):

(a)i The downward force exerted by the fan-block system (or “mass”, as read on a balance, of the system)

(a)ii A spring scale (or mechanical balance)

(a)iii Set the fan angle to 90° . With the fan off, hang the system from a spring scale to find its weight (or place the system on a mechanical balance to find its mass). Turn the fan on, and record the force from the spring scale (or the mass reading from the mechanical balance).

Example 2:

(a)i Mass of the fan-block system and low-friction cart. Velocity of the cart as a function of time.

(a)ii Scale to measure mass. Motion detector to measure velocity as a function of time.

(a)iii Place the system and a low-friction cart on a scale to find the total mass of the fan-block and cart. Next, on a horizontal surface, place the fan-block on the cart, set the fan angle to zero, turn on the fan, and release the cart from rest. Use the motion detector to graph the cart velocity as a function of time.

iv) 1 point

For correctly indicating a method to determine the force exerted by air on the fan, using the measurements and procedure from parts (a) i, ii, and iii

1 point

Example 1 (and 1a), continued from above:

Take the difference in the force readings with the fan off and the fan on. (1a: Multiply the difference in the mass readings by g). This difference is the force exerted by air on the fan.

Example 2, continued from above:

Estimate the average slope of the graph, which is the acceleration. Multiply acceleration by the total mass to find the force exerted by air on the fan.

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Question 3 (continued)

**Distribution
of points**

(b)

i, ii, and iii) 2 points

Note: Parts i, ii, and iii are read together

Note: In parts (b) i, ii, and iii, it is acceptable to say "the same as in part (a), with the following changes:", with changes to the procedure in part (a) then listed.

For describing measurements that would allow determination of the acceleration

1 point

For describing a procedure that is practical/feasible in a school laboratory

1 point

Example 3:

Place the fan-block on the track, with angle = 0. Place a motion detector behind it, set to graph velocity vs. time. Turn fan on, and release from rest. Repeat the procedure several times.

Example 4:

Place the fan-block on the track, with angle = 0. Place a photogate just in front of the block, and the second photogate a meter or two in front of the block. Measure distance from front of block to the front of that second photogate. Turn on fan and release the block from rest. The photogates measure the time it takes the block to travel the measured distance to the second photogate.

Example 5:

(b)i Angle of the fan. Velocity of fan-block system as a function of time.

(b)ii Protractor to measure the angle.

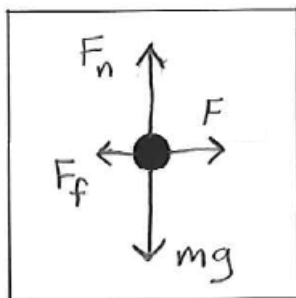
Motion detector to measure the velocity and time.

(b)iii Adjust the fan angle until the fan-block slides with a constant speed after being given a push. Use the motion detector to determine whether the speed is constant (zero slope on a graph of velocity as a function of time).

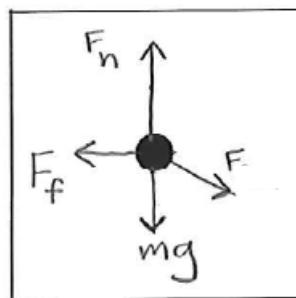
iv) 1 point

For correctly drawn and labeled vectors for the force exerted by air on fan and for the frictional force exerted on the block by the track, corresponding to the experiment described in part (b)iii. Other forces correctly drawn and labeled AND no incorrect forces, corresponding to the experiment described in part (b)iii.

1 point



Examples 3 & 4



Example 5

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Question 3 (continued)

	Distribution of points
(c) 4 points	
For using kinematic data to obtain acceleration, a , or showing that $a = 0$ Alternatively the point can be earned using the Work-Energy solution with kinetic energy $K = (1/2)mv^2$	1 point
For relating frictional force and coefficient of kinetic friction as $F_f = \mu_k F_N$	1 point
For a correct representation of the normal force, correctly referring back to a measurement of the fan-block system	1 point
$F_N = mg$ if $\theta = 0$ or $F_N = mg + F \sin \theta$ if $\theta \neq 0$	
For using Newton's second law, consistently with normal force above, for the appropriate forces or force components in the direction of the fan's motion that would lead to the determination of the coefficient of kinetic friction $F_{\text{net}} = ma$ $F - F_f = ma$	1 point
$\mu_k = \frac{F - ma}{mg}$ if $\theta = 0$ or $\mu_k = \frac{F \cos \theta - ma}{mg + F \sin \theta}$ if $\theta \neq 0$	
Alternatively the point can be earned using the Work-Energy solution that would lead to the determination of the coefficient of kinetic friction.	
Example 3 (continued from above): $\theta = 0$ Determine the slope of each v vs. t graph, to get the acceleration for each trial, and average the accelerations. Call this average acceleration a . The kinetic frictional force exerted by track on block is $F_f = \mu F_n = \mu mg$, where m is the mass of the fan-block as found in part (a), since the acceleration is purely horizontal and hence F_n and mg must balance. So, by Newton's second law, letting F denote the force exerted by the air on the fan found in part (a), we get $F_{\text{net}} = ma$ $F - F_f = ma$ $F - \mu mg = ma$	
Solve for μ to get $\mu = \frac{F - ma}{mg}$	
Example 4 (continued from above): $\theta = 0$ Let d denote distance from block to second photogate and t denote the corresponding time. Since $v_0 = 0$, $d = \frac{1}{2}at^2$ and hence $a = \frac{2d}{t^2}$. For each trial, calculate that acceleration, and take the average, which becomes our best estimate of a . [Rest of solution is as in example 3]	

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Question 3 (continued)

**Distribution
of points**

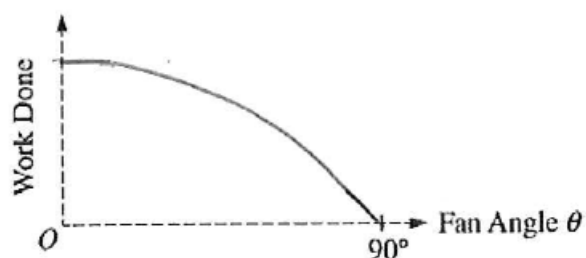
Example 5 (continued from above): $\theta = 0$ Since the acceleration is zero, the net horizontal force is zero. From Newton's second law,

$$F_f = \mu mg = F \cos \theta$$

Solving for μ_k :

$$\mu_k = \frac{F \cos \theta - ma}{mg + F \sin \theta} \text{ if } \theta \neq 0$$

(d) 2 points



A correct graph shows the work done as proportional to $\cos \theta$.

For showing a curve that continually decreases from a maximum value at 0° to a minimum value of zero at 90°

1 point

For showing a curve that is concave-downward (i.e. that has a negative second derivative)

1 point

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Question 4

7 points total

**Distribution
of points**

(a)

i. 1 point

Correct answer: “The same”

For selecting “The same” and explaining that both ladybugs have the same angular displacement over the same interval of time, or that all points on the disk rotate at the same rate 1 point

Notes: No credit is earned if the wrong answer is selected.

No credit is earned for a correct selection with a wrong explanation or no explanation given.

ii. 1 point

Correct answer: “Greater”

For selecting “Greater” and explaining that linear speed is proportional to (or increases with) distance from the center ($v = \omega R$) 1 point

Notes: No credit is earned if the wrong answer is selected.

No credit is earned for a correct selection with a wrong explanation or no explanation given.

(b) 2 points

Correct answer: “Decrease”

If the wrong answer is selected, at most one point can be earned for the explanation.

For stating or implying that the system angular momentum does not change 1 point

For stating that the ladybug gained angular momentum, the disk lost angular momentum 1 point

Note: Full credit can be earned for an alternate solution involving the force exerted by the ladybug (while accelerating) causing a torque on the disk.

(c)

i. 2 points

For using an equation expressing the conservation of angular momentum with I and ω 1 point

$$I_2\omega_2 = I_1\omega_1$$

For correctly substituting the angular speed values and obtaining a correct answer or an algebraic equivalent of the correct answer 1 point

$$I_2 = \frac{I_1\omega_1}{\omega_2} = \frac{I_1(1/2.5)}{(1/2)}$$

$$I_2 = \frac{4}{5}I_1$$

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Question 4 (continued)

**Distribution
of points**

ii. 1 point

Correct answer: “Yes”

For selecting “Yes” and for providing evidence that a torque is exerted on the disk (e.g., the disk’s angular momentum/speed changes, or the ladybug exerts a non-radial force on the disk) 1 point

Notes: No credit is earned if the wrong answer is selected.

No credit is earned for a correct selection with a wrong explanation or no explanation given.

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Question 5

7 points total

**Distribution
of points**

(a) 1 point

For an explanation using superposition

1 point

Note: The term “constructive interference”, while not a part of this course, is an acceptable explanation.

Example:

At *A*, a crest from wave 1 overlaps a crest from wave 2, and their superposition therefore results in a higher crest, as big as it can get for these two waves.

(b) 1 point

For an explanation using superposition

1 point

Note: The term “destructive interference”, while not a part of this course, is an acceptable explanation.

Example:

At *B*, a crest from wave 1 overlaps an equally deep trough from wave 2, and hence the waves “cancel out” resulting in a small amplitude.

(c) 5 points

For mentioning that waves 1 and 2 become more similar (or, frequency/period of wave 2 decreases/increases), as points *A* and *B* (loud and soft sounds) in the graph of wave 3 get further apart

1 point

For explaining why *A* and *B* (loud and soft sounds) get further apart in the graph of wave 3

1 point

Examples:

...because there are more wave peaks that constructively interfere before they start to destructively interfere.

or

... because the time interval between when the peaks coincide increases.

or

... because it takes more time for the peaks to get out of alignment.

For mentioning that *A* and *B* (loud and soft sounds) getting further apart corresponds to a longer time interval between the piano tuner hearing loud and soft sounds OR the piano tuner hears a lower beat frequency

1 point

For mentioning that, when the string frequency matches that of the fork, the time interval between the loud and soft sounds is infinite OR the loudness no longer fluctuates OR there are no more beats heard by the tuner

1 point

For a response that has sufficient paragraph structure, as described in the published requirements for the paragraph length response

1 point

Example:

The piano tuner hears alternating loud to soft to loud corresponding to *A* to *B* to the next *A* on wave 3. This alternating pattern is called “beats.” As the frequency of the sound produced by the string (wave 2) becomes closer and closer to the tuning fork frequency (wave 1), the distance on the wave 3 graph between *A* and *B* (and the next *A*) becomes greater. That distance on the graph is really the time between beats, i.e., the beat period. So, as the string frequency approaches the tuning fork frequency, the period of the beats increases. In the limit as the frequencies of waves 1 and 2 match perfectly, this period approaches infinity, which means the piano tuner no longer hears beats at all—the piano string is now perfectly tuned.